

Phase 2 Methodology Plan

Rebuilding the benchmark's foundation after MLHC 2026 rejection: clinical-annotation groundwork, a tightened rubric, and an isolated-variable pilot

Prepared by	Uyen Tran
Prepared for	Project advisor · research team (handoff continuation of Submission 175)
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Scope	Methodology and instructions only — no implementation claims; distinguishes clinical <i>annotation/validation</i> from a clinical <i>trial</i>
Milestones	Wednesday post-class check-in · Friday afternoon planning meeting

0 Framing

This phase is **clinical annotation and validation work, not a clinical trial**. We are not assigning patients to an intervention or measuring patient outcomes — we are eliciting structured clinical judgment to validate the benchmark's ground truth: which scenarios should reasonably change with a patient's age, which should not, and which need more context than age alone provides. No clinical collaborator is on the team yet, so this phase is scoped to be fully executable now, and to leave a clean, self-contained handoff packet for whoever joins.

Two outcomes anchor every decision below: (1) a resubmission-quality paper that answers the MLHC reviewers directly rather than re-arguing the same claims, and (2) preliminary methodology and data that can support an NIH R21 proposal. Both favor the same near-term move: fix the conceptual and measurement problems the reviewers actually found, before scaling anything back up.

1 What the Rejection Actually Found

The reviews (Submission 175, decision: reject) converged on one structural issue underneath the itemized complaints: **the framework treated any deviation from an age-neutral response as bias**, without a way to tell clinically appropriate age adaptation apart from an actual safety or representational harm — and the two components that would have let a reader trust the disparity numbers (the prompt-generation process and the scoring judge) were not isolated from the age variable they were supposed to measure. I traced each cited weakness to where it actually lives in the current code, since the fixes below are only credible if they target the real mechanism.

REVIEWER CONCERN	WHERE IT LIVES IN THE REPO	WHY IT UNDERMINES THE RESULT	ADDRESSED BY
Age-conditioned prompts aren't isolated to age alone (fG4L)	<code>notebooks/01_prompt_generation.ipynb</code> : an LLM rewrites each base question into 4 age variants, inventing new symptom detail, duration, and phrasing per variant	Safety/tone differences could come from the rewritten symptom framing, not the age cue — the single-variable claim in the paper isn't	Workstream D — template-based substitution

		actually true of the pipeline	
Single judge, own age-sensitivity untested (zuvy, meta-review)	<code>notebooks/04_medgemma_safety_audit.ipynb</code> : MedGemma (<code>google/medgemma-4b-it</code>) scores <i>both</i> safety and severity for every response	If the judge itself scores identical text differently when the age label changes, cross-model "agreement" is exactly what a biased shared judge would produce — this is the objection the rebuttal never resolved	Workstream C + judge calibration arm in Workstream D
Severity weights' provenance is unstated (zuvy)	<code>metrics/SafetyMetricsCalculator.py</code> hardcodes {minor:0, moderate:1, concerning:2, severe:3}; the actual per-response label is assigned by MedGemma itself via a prompt in notebook 04, not a human-applied rubric	SRS is linearly driven by these weights, but they were never independently assigned or checked — the "predefined rubric" description in the rebuttal doesn't match the implementation	Workstream C
TDI called "representational harm" but is keyword counting (zuvy, fG4L)	<code>metrics/TDICalculator.py</code> : formality score is a weighted count of contractions, hedge words, imperative verbs, and a fixed patronizing-phrase regex list	It's a real, defensible linguistic-variation statistic, but the paper's harm framing outran what it measures	Reframe in Workstream B/C; no harm claim without taxonomy-conditioned validation
Thin model coverage (fG4L)	Only GPT-5-mini and Llama-3.2-1B-Instruct evaluated	A 1B model isn't representative of deployment-scale open models	Deferred — not a methodology fix, revisit before resubmission once Workstreams A–D are solid

TWO DATA-INTEGRITY BUGS FOUND WHILE READING THE PIPELINE (NOT IN THE REVIEWS)

1. Asymmetric default-value imputation mechanically manufactures a negative ADSB. In

`SafetyMetricsCalculator.calculate_adsb()` , whenever MedGemma returns a non-numeric safety score, the *baseline* row is defaulted to `2` (safer) and the *age-conditioned* row is defaulted to `3` (less safe) — before any real comparison happens. `results/metrics/gpt.txt` shows this path was hit for **4,326 of 53,395 rows (~8.1%)**. Every one of those rows is coded as "age made it less safe" by construction, regardless of what the model actually said. This needs to be fixed — symmetric defaults, exclusion, or a retry — before any pilot number is treated as real.

2. No range validation on parsed scores. The same file logs an average safety score of `901.9441` for the GPT senior cohort — safety is a 1–5 scale. The parser only checks `str(x).isdigit()` , so any all-digit judge output is

accepted as a valid score with no bounds check, silently corrupting the mean. Needs a hard 1–5 (safety) / 0–3 (severity) assertion, and an audit of how many prior results this touched.

2 Phase 2 Workstreams

These are the four workstreams from the advisor's handoff instructions, in the order they were assigned, each translated into concrete steps against this specific codebase.

A — Scenario Pool Refinement

Goal: narrow the existing 10,679 base scenarios (deduplicated `counsel_chat` + `healthsearchqa` questions, see `data/base_prompts.csv`) into a smaller, well-characterized pool sized for hand annotation now and clinician review later — not the full sweep, which is still confounded per Workstream D.

1. Semantic (not just exact-string) de-duplication — the existing pipeline only drops literal duplicate strings.
2. Tag each scenario with a coarse clinical topic/category; none exists today beyond the physical/mental-health split, which makes any later sampling ad hoc.
3. Flag and set aside scenarios that are already inherently age-anchored in their base text (e.g. only sensible for a minor) — these can't be neutrally recombined across all four age groups.
4. Target size for the curated pilot pool: **150–300 scenarios** — enough for a meaningful split across the three taxonomy classes (Workstream B) and for two annotators to fully label before Friday, but small enough to iterate on quickly.
5. Version as `data/scenario_pool_v2.csv` with a short selection log documenting what was dropped and why. Leave the original 10,679-scenario set untouched as the record of what was submitted.

B — Age-Relevance Taxonomy

WHY THIS IS THE CENTRAL FIX

Both fG4L and the meta-review point at the same root problem: the original framework can't tell a clinically *appropriate* age adaptation from an unfair one. This taxonomy is the mechanism that fixes that — every later metric should be read conditional on which class a scenario falls into.

Age-relevant — correct clinical guidance genuinely differs by age (dosing, screening intervals, fall risk, developmental context). Age-conditioned variation in the advice is *expected*; the audit question shifts from "does it differ from baseline?" to "is the age-specific advice clinically correct?"

Age-irrelevant — age carries no clinical bearing on the correct answer (e.g. general disease-mechanism or first-aid questions). This is the class where ADSB / SRS / TDI, as originally conceived, apply cleanly — any safety or tone shift correlated with age here is a genuine bias signal.

Insufficient-context — age alone can't determine the right advice without more clinical detail (comorbidities, severity, medication history). The "correct" model behavior may be to ask a clarifying question rather than assert a directive answer; audit for premature certainty, not just a safety score.

1. Draft the three-way coding scheme above into a standalone document with 2–3 worked examples per class, pulled from the curated pool (Workstream A).
2. Have at least two people independently label a sample of the curated pool against the taxonomy — not a single annotator, which was the gap reviewers flagged in the original submission.
3. Compute inter-rater agreement (Cohen's κ) on the pilot batch, even informally. Tracking this from day one is itself evidence of rigor worth citing in both the resubmission and an eventual R21 preliminary-work section.
4. Package the taxonomy plus disagreement cases as the first artifact a clinical collaborator reviews once identified.

C — Rating Rubric Tightening

Today, both the safety scale (1–5) and severity weights (0–3) exist only as natural-language instructions fed to MedGemma, and MedGemma assigns severity to its own safety call — a single-source, circular design that reviewers correctly flagged as unverifiable.

1. Externalize the rubric out of the LLM instruction string into a standalone, versioned rubric document, independent of any one model or scoring run.
2. Add worked examples for each of the 5 safety levels and 4 severity levels, anchored in the taxonomy (Workstream B) — e.g. a severity-2 example in an age-irrelevant scenario looks different from one in an age-relevant scenario.
3. Decouple severity from safety: severity should describe the clinical consequence of the scenario category itself (largely invariant to age or to which model answered), not be re-derived from the same model's own safety judgment. This removes the circularity in the current design.
4. Fold in the two data-integrity fixes from Section 1 — symmetric/valid score handling is part of what "tightening the rubric" has to mean operationally, not just prose changes.

D — Controlled Computational Pilot

This is the load-bearing piece: the direct answer to fG4L's confound objection, and the first result worth bringing back to the advisor.

- **Design:** replace the LLM-rewrite step (notebook 01) with slot-based templates. Hold each scenario's symptoms, duration, and question fixed; vary only an explicit age slot ("I am a 16-year-old..." / "I am a 45-year-old..." / age omitted for baseline) inside one fixed sentence frame, so age is the only thing that changes across the five conditions.
- **Scope:** small and fast — 30–50 scenarios from the curated pool $\times$ 5 conditions (baseline + 4 age groups) $\times$ one model, sized to run in an afternoon and produce a number to bring to Friday, not the full sweep.
- **Judge-calibration arm (same pilot):** for a subsample, re-score the *identical* generated response after swapping only its age label before it reaches MedGemma. If the safety or severity score shifts on identical text purely because the label changed, that isolates and quantifies scorer bias directly — the exact test the rebuttal was missing.
- **Deliverable:** a 2–3 page internal memo covering (i) whether template-only substitution reproduces the original disparities, (ii) whether the judge shows label-driven drift, (iii) recommended next steps.

3 Data-Integrity Checklist

Do this before re-running any metric on new data — otherwise Workstream D's pilot inherits the same silent bugs.

- ☐ Fix the asymmetric default-value imputation in `calculate_adsb()` (baseline→2, age→3 on parse failure); exclude unparseable rows or use a symmetric/retry policy instead, and report the affected percentage either way.
- ☐ Add a hard valid-range check (1–5 safety, 0–3 severity) so a malformed judge output is flagged and dropped rather than silently entering an aggregate mean.
- ☐ Persist the parse-failure rate per model / age-group as a saved QA metric on every run, not just a console print.
- ☐ Record and pin the exact judge checkpoint (confirmed in code: `google/medgemma-4b-it`) and version/commit used for the pilot — directly answers zuvy's "4B or 27B?" question.

4 Clinical-Collaborator Handoff Packet

Build this in parallel with the advisor's outreach, so a clinical collaborator can validate on arrival instead of starting from scratch:

1. Taxonomy document with worked examples (Workstream B).
2. Externalized rubric with worked examples (Workstream C).
3. Curated scenario pool (Workstream A) with taxonomy pre-labels and flagged disagreements.
4. A no-code annotation sheet (a plain spreadsheet: scenario, current label, agree/disagree, comment) — no need for a custom interface at this stage.
5. The pilot memo (Workstream D), so the collaborator sees concrete examples of "age-relevant adaptation" versus "bias" rather than the abstract definitions alone.

5 Timeline

Now → Wednesday (after-class check-in)

Finish the first-pass taxonomy draft (B) and start scenario pool refinement (A); bring open taxonomy edge cases to discuss.

Wednesday → Friday

Draft rubric v2 (C), fix the two data-integrity bugs (Section 3), start the template-substitution pilot script (D).

Friday afternoon meeting

Review taxonomy and rubric drafts together; look at early pilot numbers if ready; agree on curated-pool size; align on what "done" looks like for this phase before clinician review begins.

After Friday

Run the full pilot including the judge-calibration arm; assemble the handoff packet (Section 4) alongside the advisor's clinical-collaborator outreach.

6 Open Questions for Friday

- **Taxonomy edge cases:** how do we classify a scenario that is clinically age-irrelevant but where health literacy and phrasing genuinely differ by age — literacy and clinical relevance seem like two different axes that the three-way scheme may need to keep separate.
- **Pilot model choice:** keep GPT-5-mini + Llama-3.2-1B for continuity with the rejected submission, or add a larger open-weight model now, given fG4L's "1B isn't representative" critique?
- **Curated pool size:** does 150–300 scenarios give enough power once split three ways by taxonomy class, especially for the smallest class?
- **Publication strategy:** does the reframed contribution (taxonomy + validated rubric + isolated-variable pilot) still fit the same venue's next cycle, or does the narrower, more rigorous scope point somewhere else?

SOURCE REFERENCES FOR THIS MEMO

Diagnosis in Sections 1 and 3 is grounded directly in this repository: `notebooks/01_prompt_generation.ipynb` (prompt-variant generation), `notebooks/04_medgemma_safety_audit.ipynb` (safety/severity scoring), `metrics/SafetyMetricsCalculator.py` (ADSB/SRS, including the imputation bug), `metrics/TDICalculator.py` (tone metric), and `results/metrics/gpt.txt` (the run output showing the 4,326-row imputation count and the 901.9441 outlier). Reviewer quotes are drawn from the MLHC 2026 Submission 175 decision thread (Reviewers zuvy, rfZa, fG4L; Area Chair meta-review).